

MATH 189 HW 2

Dani Nguyen

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Question 1

- a. Since IQ and GPA is fixed, we will denote those values as a constant C

$$\begin{aligned} Y &= C + 35X_3 - 10X_5 \\ &= C + \mathbb{I}\{\text{has college}\}(35 - 10X_5) \end{aligned}$$

When fixing IQ and GPA, the model predicts that college graduates will earn less if their GPA is higher than 3.5. Therefore, option iii. is correct.

- b. For a college graduate with IQ 110 and GPA 4.0,

$$\begin{aligned} Y &= 50 + 20(4.0) + 0.07(110) + 35(1) + 0.01(4.0 \cdot 110) - 10(4.0 \cdot 1) \\ &= 137.1 \text{ thousand dollars} \end{aligned}$$

- c. False, because GPA and IQ are on average very large numbers, especially when multiplied, that even a low coefficient can lead to large effects in salary. This does not mean that there is little evidence of an interaction effect, one would need to look at the p -values of the interaction term to know whether there exists an interaction effect.

Question 2

$$\begin{aligned}
 \hat{y}_i &= x_i \hat{\beta} \\
 &= x_i \frac{\sum_{i=1}^n x_i y_i}{\sum_{i'=1}^n x_{i'}^2} \\
 &= \frac{\sum_{i=1}^n x_i^2 y_i}{\sum_{i'=1}^n x_{i'}^2} \\
 &= \sum_{i=1}^n \frac{x_i^2}{\sum_{i'=1}^n x_{i'}^2} y_i
 \end{aligned}$$

Let $a_i = \frac{x_i^2}{\sum_{i'=1}^n x_{i'}^2}$

$$\hat{y}_i = \sum_{i=1}^n a_i y_i$$

Question 3

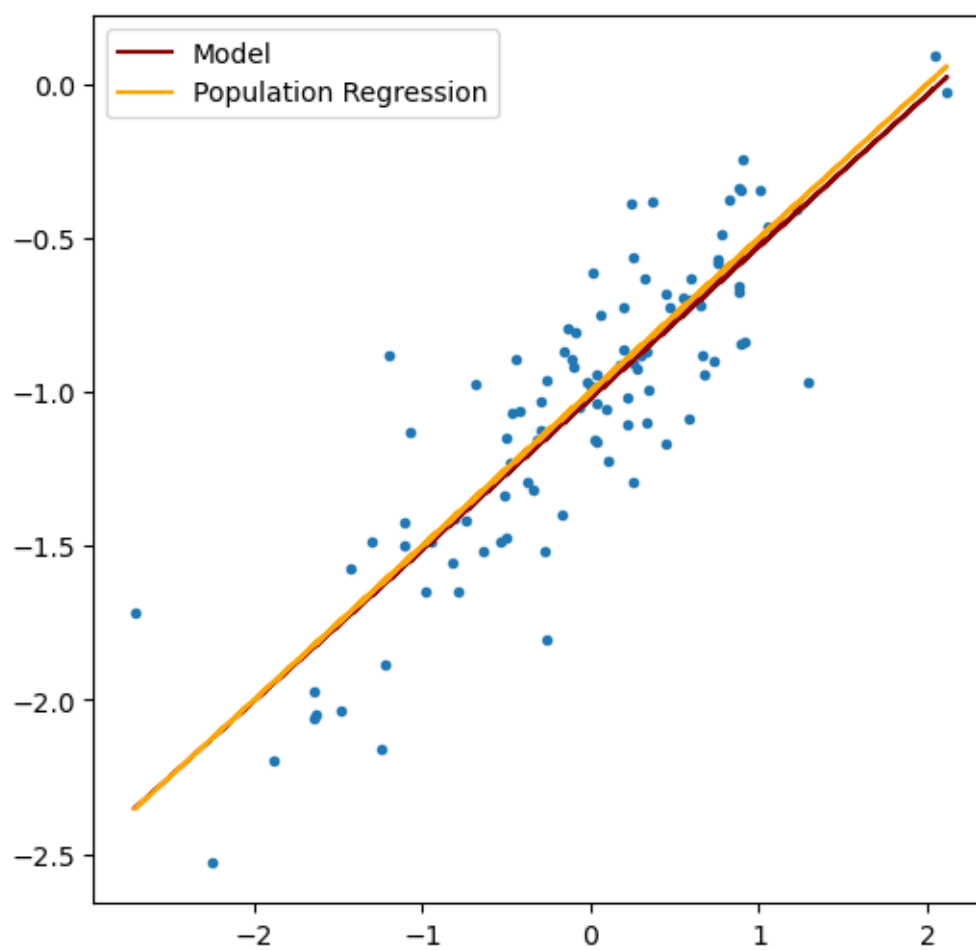
Recall that $\beta_0 = \bar{y} - \beta_1 \bar{x}$. This can be rearranged to $\bar{y} = \beta_0 + \beta_1 \bar{x}$. This means that the equation for simple linear regression is centered at and will always pass through the point $(\bar{x}, \bar{y})$.

Question 4

$$\begin{aligned}
 R^2 &= \frac{TSS - RSS}{TSS} \\
 &= \frac{\sum_{i=1}^n (y_i)^2 - \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2}{\sum_{i=1}^n (y_i)^2} \\
 &= \frac{\sum_{i=1}^n (y_i)^2 - \sum_{i=1}^n (y_i)^2 + \left[\frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} \right]^2}{\sum_{i=1}^n (y_i)^2} \\
 &= \frac{\sum_{i=1}^n (x_i y_i)^2}{\sum_{i=1}^n (x_i)^2 \sum_{i=1}^n (y_i)^2} \\
 R &= \frac{\sum_{i=1}^n (x_i y_i)}{\sum_{i=1}^n (x_i) \sum_{i=1}^n (y_i)} = \frac{Cov(X_i Y_i)}{Var(X_i)^{\frac{1}{2}} Var(Y_i)^{\frac{1}{2}}}
 \end{aligned}$$

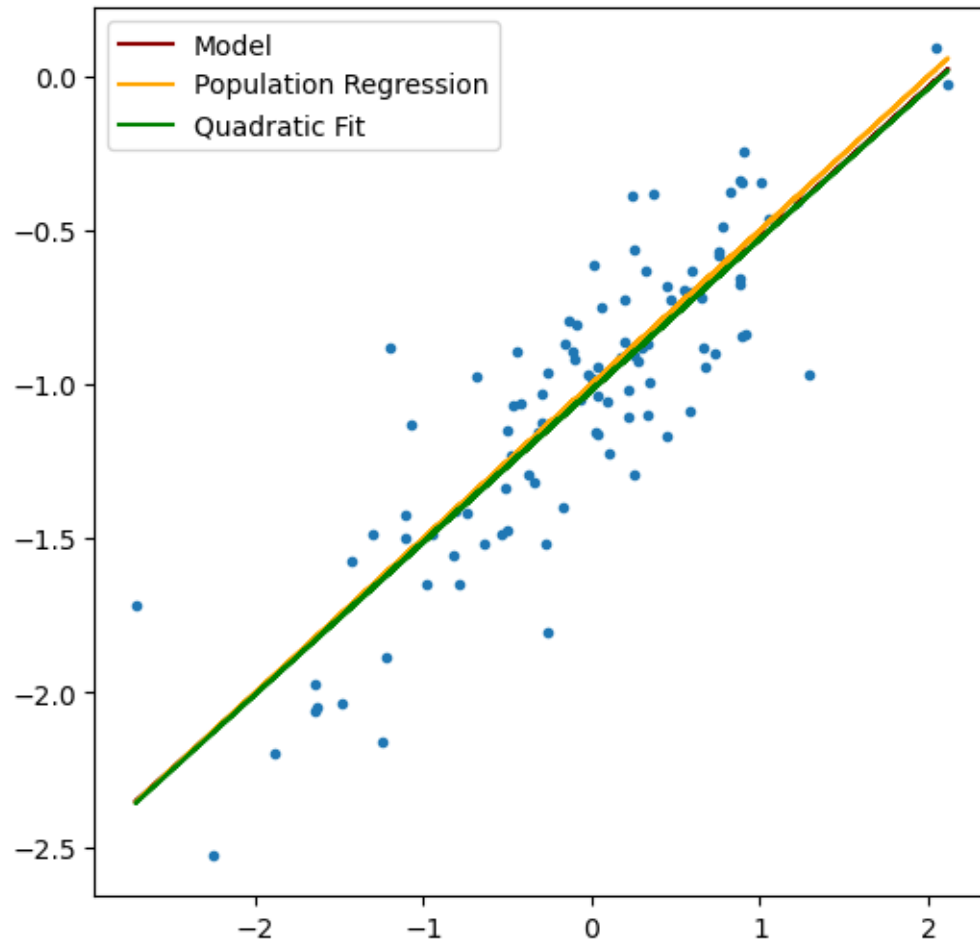
Question 5

- c. Length of the vector y is 11.62. In this linear model β_0 is -1 and β_1 is 0.5 .
- d. The distribution is linear, positive, and moderately spread out.
- e. The model is linear and has positive slope, $\hat{\beta}_1$ is 0.49214551 and $\hat{\beta}_0$ is -1.01900639 which is fairly close to our β_1 of 0.5 and β_0 of -1

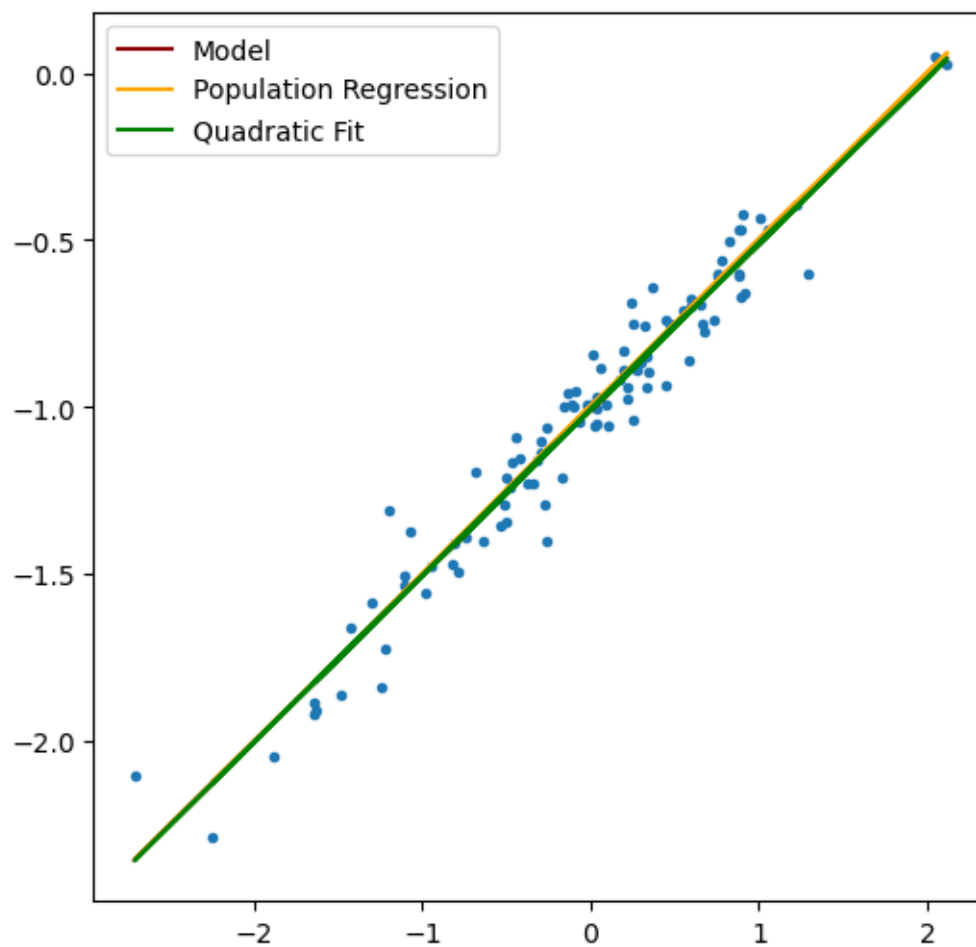


f.

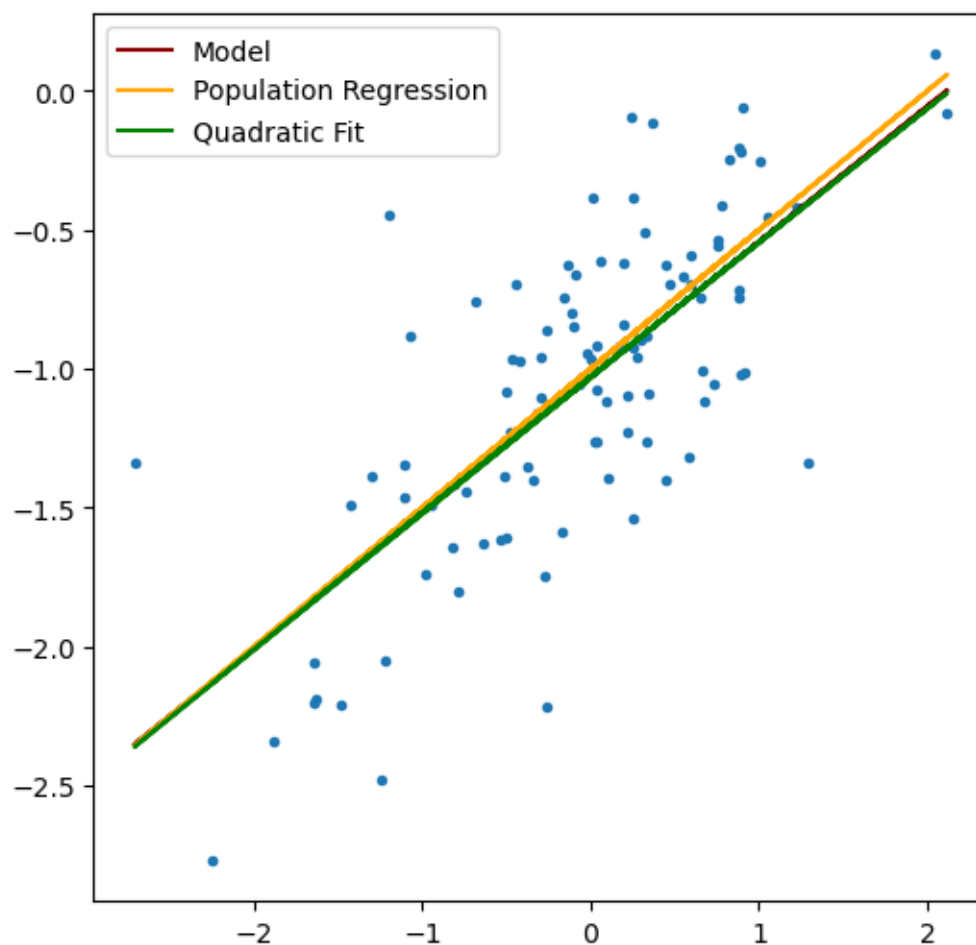
- g. The quadratic line of best fit seems to fit just as well as the linear model, although there is no evidence that it improves the fit.



- h. The points are much closer to the least squares regression line, and the differences between the model and the population regression decreases



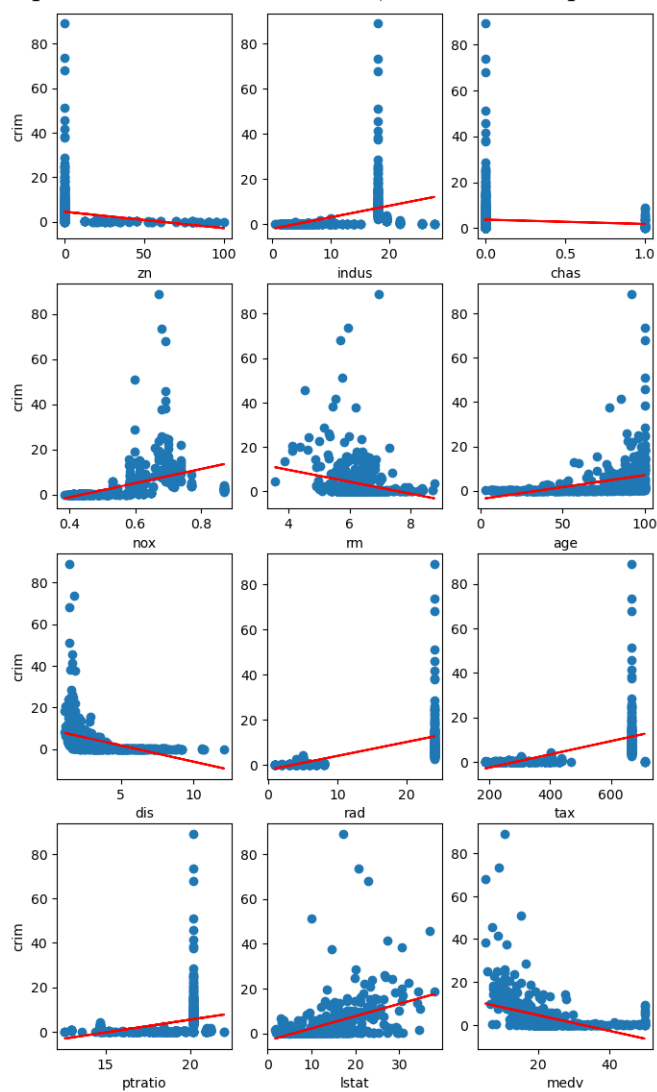
- i. The points are more spread out along the least squares regression line, and the differences between the model and the population regression increases



- j. For the noisy dataset, the confidence interval for β_0 is $[-1.03 \pm 0.0798]$, β_1 is $[0.4874 \pm 0.0933]$.
 For the original dataset the confidence interval for β_0 is $[-1.02 \pm 0.0499]$, β_1 is $[0.4921 \pm 0.0584]$.
 For the less noisy dataset the confidence interval for β_0 is $[-1.007 \pm 0.0200]$, β_1 is $[0.4969 \pm 0.0233]$.

Question 6

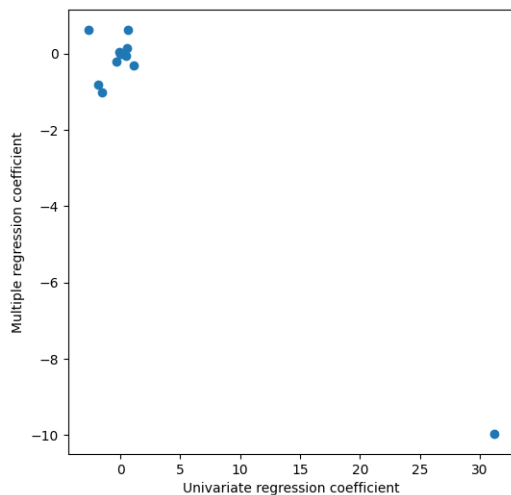
- a. None of the distributions seem to fit a linear regression model, but all of the predictors have p-value below 0.05 when fitted with crime rate per capita data aside from chas, which has a p-value of 0.209



	coef	std err	t	P> t
intercept	13.7784	7.082	1.946	0.052
zn	0.0457	0.019	2.433	0.015
indus	-0.0584	0.084	-0.698	0.486
chas	-0.8254	1.183	-0.697	0.486
nox	-9.9576	5.290	-1.882	0.060
rm	0.6289	0.607	1.036	0.301
age	-0.0008	0.018	-0.047	0.962
dis	-1.0122	0.282	-3.584	0.000
rad	0.6125	0.088	6.997	0.000
tax	-0.0038	0.005	-0.730	0.466
ptratio	-0.3041	0.186	-1.632	0.103
lstat	0.1388	0.076	1.833	0.067
medv	-0.2201	0.060	-3.678	0.000

Table 1: OLS Regression Results for Boston Crime Data

- b. We can reject the null hypothesis $H_0 = \beta_j = 0$ for zn, dis, rad, and medv as they have $P > |t|$ less than 0.05.
- c. The β_j values are mostly similar, aside from nox, which is wildly different between multivariate (-9.9576) and simple linear regression (31.2485)



d.